

1x1 - X-ARM-VLA - 50 size dataset

⑩ ID Cup / ID Machine

→ order

WB/K ✓ T	O/K X T	Bk/K X S	R/K X T
WB/Bk X ¹ S	O/Bk X ² T	Bk/Bk X ² S	R/Bk X ² S
WB/T X ² T	O/T X ² T	Bk/T X ² S	R/T X ¹ S
WB/W X ² T	O/W X ² S	Bk/W X ² T	R/W X ³ T

⑫ ID Cup / OD Machine

→ order

WB/R X ² T	O/R X ² T	Bk/R X ² S	R/R X ² T
WB/TS ✓ T	O/TS X ² S	Bk/TS X ⁴ S	R/TS X ² S
WB/Bl X ⁵ T	O/Bl X ² S	Bk/Bl X ⁴ S	R/Bl X ² T

⑩ OD Cup / ID Machine

→ order

Br/K X ² T	WC/K ✓ S	Gr/K X ² T	P/K ✓ S
Br/Bk X ² T	WC/Bk X ² T	Gr/Bk X ² T	P/Bk X ² S
Br/T X ² S	WC/T X ² S	Gr/T X ^{1,2} S	P/T X ¹⁰ T
Br/W X ¹ S	WC/W ✓ ⁹ T	Gr/W X ² T	P/W X ¹ T

⑫ OD Cup / OD Machine

→ order

Br/R X ² S	WC/R ✓ T	Gr/R X ² T	P/R X ² T
Br/TS X ² S	WC/TS X ² S	Gr/TS X ² T	P/TS X ⁵ S
Br/Bl X ² S	WC/Bl X ⁶ S	Gr/Bl X ² S	P/Bl ✓ T

1. Spilled the cup
2. Failed to grasp (hit cup, grasped air)
3. Hit the coffee machine
4. Good pose, high z
5. Wrong grasp
6. Torque sensor error

7. Moved machine back
8. Handle hooked on gripper
9. Balancing on edge
10. Dropped cup elsewhere

4x4 - X-ARM-VLA - 50 size dataset

⑩ ID Cup / ID Machine

→ order

WB/K	X ² S	O/K	X ² S	Bk/K	X ¹ S	R/K	✓S
WB/Bk	X ² S	O/Bk	X ¹ S	Bk/Bk	✓S	R/Bk	X ¹ S
WB/T	✓S	O/T	✓S	Bk/T	X ¹ S	R/T	X ¹ S
WB/W	✓S	O/W	✓S	Bk/W	✓T	R/W	X ² T

⑫ ID Cup / OD Machine

→ order

WB/R	✓S	O/R	✓T	Bk/R	X ² S	R/R	✓S
WB/TS	X ¹ S	O/TS	✓S	Bk/TS	✓S	R/TS	✓S
WB/Bl	X ² S	O/Bl	X ¹ S	Bk/Bl	✓S	R/Bl	X ² S

⑩ OD Cup / ID Machine

→ order

Br/K	X ¹ S	WC/K	✓S	Gr/K	X ⁵ S	P/K	X ⁶ S
Br/Bk	X ⁶ T	WC/Bk	X ² S	Gr/Bk	X ² T	P/Bk	X ¹ S
Br/T	X ² T	WC/T	X ² S	Gr/T	X ² S	P/T	X ² S
Br/W	X ² S	WC/W	✓S	Gr/W	X ² T	P/W	X ¹ T

⑫ OD Cup / OD Machine

→ order

Br/R	✓ ⁷ S	WC/R	✓ ⁷ S	Gr/R	X ² S	P/R	✓S
Br/TS	X ² S	WC/TS	X ^{7,8} S	Gr/TS	X ² T	P/TS	X ^{1,8} S
Br/Bl	✓S	WC/Bl	X ^{2,6} S	Gr/Bl	✓ ^{7,9} T	P/Bl	✓S

1. Spilled the cup
2. Failed to grasp (hit cup, grasped air)
3. Hit the coffee machine
4. Good pose, high z
5. Wrong grasp
6. Torque sensor error

7. Moved machine back
8. Handle hooked on gripper
9. Balancing on edge

1x1 - X-ARM - VLA - 100 size dataset

16) ID Cup / ID Machine

→ order

WB / K	O / K	Bk / K	R / K
WB / Bk	O / Bk	Bk / Bk	R / Bk
WB / T	O / T	Bk / T	R / T
WB / W	O / W	Bk / W	R / W

12) ID Cup / OD Machine

→ order

WB / R	O / R	Bk / R	R / R
WB / TS	O / TS	Bk / TS	R / TS
WB / Bl	O / Bl	Bk / Bl	R / Bl

16) OD Cup / ID Machine

→ order

Br / K	WC / K	Gr / K	P / K
Br / Bk	WC / Bk	Gr / Bk	P / Bk
Br / T	WC / T	Gr / T	P / T
Br / W	WC / W	Gr / W	P / W

12) OD Cup / OD Machine

→ order

Br / R	WC / R	Gr / R	P / R
Br / TS	WC / TS	Gr / TS	P / TS
Br / Bl	WC / Bl	Gr / Bl	P / Bl

4x4 - X-ARM-VLA - 100 size dataset

⑩ ID Cup / ID Machine

→ order

WB / K	O / K
WB / Bk	O / Bk
WB / T	O / T
WB / W	O / W

Bk / K	R / K
Bk / Bk	R / Bk
Bk / T	R / T
Bk / W	R / W

⑫ ID Cup / OD Machine

→ order

WB / R	O / R
WB / TS	O / TS
WB / Bl	O / Bl

Bk / R	R / R
Bk / TS	R / TS
Bk / Bl	R / Bl

⑩ OD Cup / ID Machine

→ order

Br / K	WC / K
Br / Bk	WC / Bk
Br / T	WC / T
Br / W	WC / W

Gr / K	P / K
Gr / Bk	P / Bk
Gr / T	P / T
Gr / W	P / W

⑫ OD Cup / OD Machine

→ order

Br / R	WC / R
Br / TS	WC / TS
Br / Bl	WC / Bl

Gr / R	P / R
Gr / TS	P / TS
Gr / Bl	P / Bl

1x1 - X-ARM - VLA - ISO size dataset

(16) ID Cup / ID Machine → order

WB / K X T	O / K X" T	Bk / K X T	R / K X" T
WB / Bk X ⁸ S	O / Bk X" T	Bk / Bk X T	R / Bk X" T
WB / T X ^{1,7} T	O / T X T	Bk / T X T	R / T X ⁷ T
WB / W X ^{1,8} T	O / W X" T	Bk / W X T	R / W X" S

(12) ID Cup / OD Machine → order

WB / R ✓ ¹ T	O / R X S	Bk / R X" S	R / R X ⁷ S
WB / TS ✓ S	O / TS X S	Bk / TS X" T	R / TS X ⁶ T
WB / Bl ✓ ¹ S	O / Bl X ⁵ T	Bk / Bl X" T	R / Bl X ⁷ T

(16) OD Cup / ID Machine → order

Br / K X" T	WC / K ✓ S	Gr / K X T	P / K X ^{7,8} S
Br / Bk X ¹ S	WC / Bk X S	Gr / Bk X T	P / Bk X S
Br / T X S	WC / T X ⁷ T	Gr / T X" S	P / T X ⁷ S
Br / W X T	WC / W X T	Gr / W X T	P / W X ¹⁰ S

(12) OD Cup / OD Machine → order

Br / R X" T	WC / R X S	Gr / R X ⁷ S	P / R X ^{7,9} T
Br / TS X S	WC / TS ✓ S	Gr / TS X S	P / TS X T
Br / Bl ✓ ^{3,5} S	WC / Bl X ⁷ T	Gr / Bl X" S	P / Bl X ^{5,7} T

1. Pushed machine back
2. Tilted placement and contact on plate
3. Took double time
4. Pushed cup while trying to grasp
5. Dropped cup on edge of platform.
6. Self collision

7. Spilled cup
8. Force-torque trigger
9. Grasped inside cup
10. Gripper hooked on handle
11. Cannot lift mug

1x1 - X-ARM - VLA - WB redo

(16) ID Cup / ID Machine → order

WB / K X S ✓ S

WB / Bk X S

WB / T ✓² T

WB / W X^{1,7,10} T

(12) ID Cup / OD Machine → order

WB / R ✓¹ S

WB / TS ✓¹ S

WB / Bl X^{7,9} T

4x4 - X-ARM-VLA - ISO size dataset

(16) ID Cup / ID Machine

→ order

WB/K	✓	S	O/K	X	S	Bk/K	✓	S	R/K	✓	T
WB/Bk	✓	S	O/Bk	✓	S	Bk/Bk	X	T	R/Bk	✓ ¹	S
WB/T	✓ ¹	T	O/T	✓ ³	T	Bk/T	✓ ¹	T	R/T	X	S
WB/W	✓ ⁴	T	O/W	✓	T	Bk/W	✓ ⁴	T	R/W	X	T

(12) ID Cup / OD Machine

→ order

WB/R	✓	S	O/R	✓ ¹	T	Bk/R	✓ ^{1,5}	T	R/R	X ⁶	S
WB/TS	✓ ⁴	S	O/TS	X	S	Bk/TS	X ⁷	S	R/TS	X	S
WB/Bl	✓ ¹	S	O/Bl	✓ ¹	S	Bk/Bl	✓	T	R/Bl	✓ ²	S

(16) OD Cup / ID Machine

→ order

Br/K	✓ ²	S	WC/K	✓	S	Gr/K	✓	S	P/K	✓	S
Br/Bk	X ^{8,9}	T	WC/Bk	✓	S	Gr/Bk	✓	S	P/Bk	X	S
Br/T	✓ ^{1,2,8}	T	WC/T	X ¹⁰	S	Gr/T	X ^{1,9}	T	P/T	✓	S
Br/W	✓ ²	S	WC/W	X ⁴	T	Gr/W	✓ ²	T	P/W	X ⁶	S

(12) OD Cup / OD Machine

→ order

Br/R	X	S	WC/R	X	S	Gr/R	X	T	P/R	✓	S
Br/TS	✓ ¹	S	WC/TS	X	S	Gr/TS	✓ ²	S	P/TS	X	S
Br/Bl	X ^{1,7,8}	S	WC/Bl	X ⁸	S	Gr/Bl	✓ ¹	S	P/Bl	✓	S

1. Pushed machine back
2. Tilted placement and contact on plate
3. Took double time
4. Pushed cup while trying to grasp
5. Dropped cup on edge of platform.
6. Self collision

7. Spilled cup
8. Force-torque trigger
9. Grasped inside cup
10. Gripper hooked on handle

4x4 - X-ARM-IDP

⑩ ID Cup / ID Machine

WB / K

O / K

Bk / K

R / K

WB / Bk

O / Bk

Bk / Bk

R / Bk

WB / T

O / T

Bk / T

R / T

WB / W

O / W

Bk / W

R / W

⑫ ID Cup / OD Machine

WB / R

O / R

Bk / R

R / R

WB / TS

O / TS

Bk / TS

R / TS

WB / Bl

O / Bl

Bk / Bl

R / Bl

⑩ OD Cup / ID Machine

Br / K

WC / K

Gr / K

P / K

Br / Bk

WC / Bk

Gr / Bk

P / Bk

Br / T

WC / T

Gr / T

P / T

Br / W

WC / W

Gr / W

P / W

⑫ OD Cup / OD Machine

Br / R

WC / R

Gr / R

P / R

Br / TS

WC / TS

Gr / TS

P / TS

Br / Bl

WC / Bl

Gr / Bl

P / Bl

1x1 - X-ARM - R2R2R

16 ID Cup / ID Machine

WB / K	O / K	Bk / K	R / K
WB / Bk	O / Bk	Bk / Bk	R / Bk
WB / T	O / T	Bk / T	R / T
WB / W	O / W	Bk / W	R / W

12 ID Cup / OD Machine

WB / R	O / R	Bk / R	R / R
WB / TS	O / TS	Bk / TS	R / TS
WB / Bl	O / Bl	Bk / Bl	R / Bl

16 OD Cup / ID Machine

Br / K	WC / K	Gr / K	P / K
Br / Bk	WC / Bk	Gr / Bk	P / Bk
Br / T	WC / T	Gr / T	P / T
Br / W	WC / W	Gr / W	P / W

12 OD Cup / OD Machine

Br / R	WC / R	Gr / R	P / R
Br / TS	WC / TS	Gr / TS	P / TS
Br / Bl	WC / Bl	Gr / Bl	P / Bl

4x4 - X-ARM-R2R2R

⑩ ID Cup / ID Machine

WB / K	O / K	Bk / K	R / K
WB / Bk	O / Bk	Bk / Bk	R / Bk
WB / T	O / T	Bk / T	R / T
WB / W	O / W	Bk / W	R / W

⑫ ID Cup / OD Machine

WB / R	O / R	Bk / R	R / R
WB / TS	O / TS	Bk / TS	R / TS
WB / Bl	O / Bl	Bk / Bl	R / Bl

⑩ OD Cup / ID Machine

Br / K	WC / K	Gr / K	P / K
Br / Bk	WC / Bk	Gr / Bk	P / Bk
Br / T	WC / T	Gr / T	P / T
Br / W	WC / W	Gr / W	P / W

⑫ OD Cup / OD Machine

Br / R	WC / R	Gr / R	P / R
Br / TS	WC / TS	Gr / TS	P / TS
Br / Bl	WC / Bl	Gr / Bl	P / Bl

IX1 - X-ARM-SAP

⑩ ID Cup / ID Machine

WB / K	O / K	Bk / K	R / K
WB / Bk	O / Bk	Bk / Bk	R / Bk
WB / T	O / T	Bk / T	R / T
WB / W	O / W	Bk / W	R / W

⑩ ID Cup / OD Machine

WB / R	O / R	Bk / R	R / R
WB / Bl	O / Bl	Bk / Bl	R / Bl
WB / TS	O / TS	Bk / TS	R / TS

⑩ ID Cup / ID Machine

Br / K	WC / K	Gr / K	P / K
Br / Bk	WC / Bk	Gr / Bk	P / Bk
Br / T	WC / T	Gr / T	P / T
Br / W	WC / W	Gr / W	P / W

⑩ ID Cup / OD Machine

Br / R	WC / R	Gr / R	P / R
Br / Bl	WC / Bl	Gr / Bl	P / Bl
Br / TS	WC / TS	Gr / TS	P / TS

4x4 - X-ARM - SAP

(16) ID Cup / ID Machine

WB / K	O / K	Bk / K	R / K
WB / Bk	O / Bk	Bk / Bk	R / Bk
WB / T	O / T	Bk / T	R / T
WB / W	O / W	Bk / W	R / W

(12) ID Cup / OD Machine

WB / R	O / R	Bk / R	R / R
WB / TS	O / TS	Bk / TS	R / TS
WB / Bl	O / Bl	Bk / Bl	R / Bl

(16) OD Cup / ID Machine

Br / K	WC / K	Gr / K	P / K
Br / Bk	WC / Bk	Gr / Bk	P / Bk
Br / T	WC / T	Gr / T	P / T
Br / W	WC / W	Gr / W	P / W

(12) OD Cup / OD Machine

Br / R	WC / R	Gr / R	P / R
Br / TS	WC / TS	Gr / TS	P / TS
Br / Bl	WC / Bl	Gr / Bl	P / Bl

4x4 - X-ARM-VLA

OLD - ISO size dataset

(16) ID Cup / ID Machine → order

WB / K ✓ T	O / K ✓ S	Bk / K X T	R / K ✓ T
WB / Bk ✓ S	O / Bk ✓ T	Bk / Bk ✓ S	R / Bk ✓ T
WB / T ✓ S	O / T ✓ S	Bk / T ✓ S	R / T ✓ T
WB / W ✓ S	O / W ✓ T	Bk / W ✓ T	R / W ✓ T

(12) ID Cup / OD Machine → order

WB / R ✓ S	O / R ✓ T	Bk / R ✓ T	R / R ✓ T
WB / TS X T	O / TS ✓ S	Bk / TS ✓ T	R / TS ✓ T
WB / Bl	O / Bl	Bk / Bl	R / Bl

(16) OD Cup / ID Machine → order

Br / K ✓ S	WC / K O T	Gr / K ✓ T	P / K ✓ S
Br / Bk ✓ T	WC / Bk ✓ S	Gr / Bk ✓ T	P / Bk ✓ S
Br / T ✓ S	WC / T ✓ S	Gr / T ✓ T	P / T ✓ S
Br / W ✓ T	WC / W ✓ T	Gr / W ✓ T	P / W ✓ S

O = Completed but took > 1 min

(12) OD Cup / OD Machine → order

Br / R ✓ T	WC / R ✓ [*] S	Gr / R ✓ T	P / R ✓ S
Br / TS ✓ T	WC / TS ✓ ^{**} S	Gr / TS ✓ T	P / TS ✓ S
Br / Bl	WC / Bl	Gr / Bl	P / Bl

* Not fully in cup holder

** Released while tilted